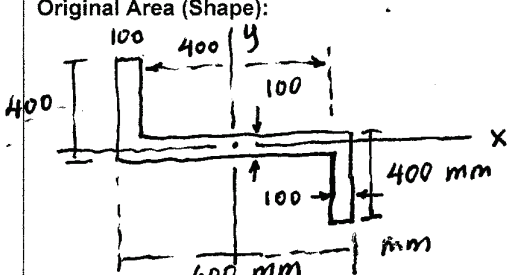
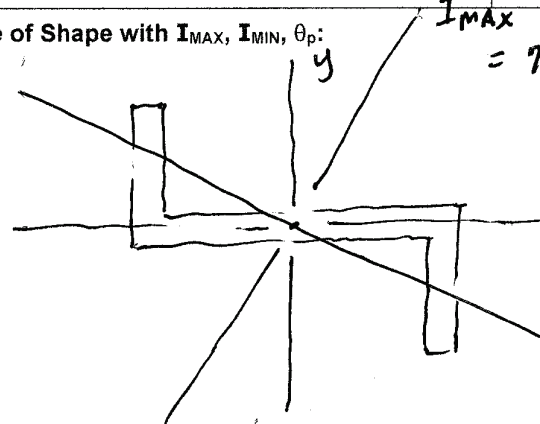
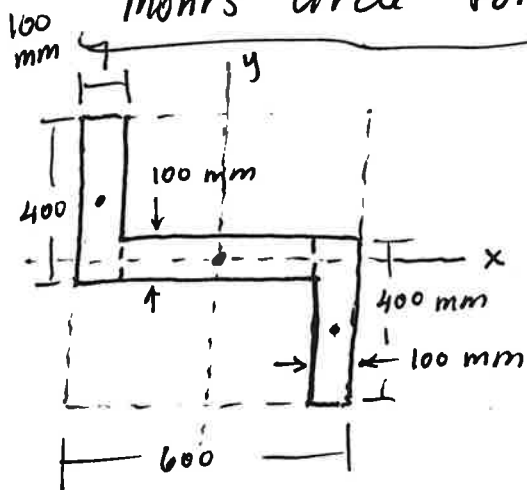


Hibbeler Ex. Prob 10.9

Worksheet: MOI Transformation (finding Principals I_{MAX} and I_{MIN}) using Equations

Step/Equations	Inputs/Calculations
Original Area (Shape): 	$I_x = 2.90 \cdot 10^9 \text{ mm}^4$ $I_y = 5.60 \cdot 10^9 \text{ mm}^4$ $I_{xy} = -3.0 \cdot 10^9 \text{ mm}^4$ (See next page for calcs)
Intermediate Calculations: $I_{AVG} = MC \text{ Center} = \left(\frac{I_x + I_y}{2} \right)$ $\left(\frac{I_x - I_y}{2} \right) =$ $\text{Radius } R = I_{xyMAX} = \sqrt{\left(\frac{I_x - I_y}{2} \right)^2 + I_{xy}^2}$	$I_{AVG} = \frac{2.9 + 5.6}{2} = 4.25 \cdot 10^9 \text{ mm}^4$ $\frac{I_x - I_y}{2} = \frac{-2.7}{2} = -1.35 \cdot 10^9$ $R = \sqrt{(1.35)^2 + (3)^2} = 3.29 \text{ mm}^4$
Principal MOI's: $I_{MAX}, I_{MIN} = I_{AVG} \mp R$ $I_{MAX}/MIN = 4.25 \pm 3.29$	$I_{MAX} = 7.54 \cdot 10^9 \text{ mm}^4$ $I_{MIN} = 0.96 \cdot 10^9 \text{ mm}^4$
Principal Angles: (Also compute $2\theta_p...$) $\theta_p = \frac{1}{2} \tan^{-1} \left(\frac{-I_{xy}}{\left(\frac{I_x - I_y}{2} \right)} \right)$	$\theta_p = \frac{1}{2} \tan^{-1} \left(\frac{-(-3)}{-1.35} \right)$ $\theta_p = -32.9^\circ \quad 2\theta = -65.8$
Which I_{MAX}, I_{MIN} is at θ_p? $I_\theta = \left(\frac{I_x + I_y}{2} \right) + \left(\frac{I_x - I_y}{2} \right) \cos(2\theta) - I_{xy} \sin(2\theta)$ $I_\theta = 4.25 + (-1.35) \cos(-65.8) - (-3) \sin(-65.8)$	$I_\theta = 0.96 \cdot 10^9 \text{ mm}^4 = I_{MIN}$ at $\theta_p = -32.9$ from x $\nearrow +\theta$ $\searrow -\theta$
Final Picture of Shape with $I_{MAX}, I_{MIN}, \theta_p$: 	$I_{MAX} = 7.54 \cdot 10^9 \text{ mm}^4$ $\theta_p = -32.9 \text{ from } x$ $I_{MIN} = 0.96 \cdot 10^9 \text{ mm}^4$

Mohr's Circle for Area MOI (See Hibbeler Ex Prob 10.9)



$$I_x = \sum (\bar{I} + Ad^2) = \frac{1}{12} (400)(100)^3 + 2 \left[\frac{1}{12} (100)(400)^3 + 100(400)(150)^2 \right]$$

$$I_x = 33.33 \cdot 10^6 + 2(1.433 \cdot 10^9)$$

$$I_x = 2.90 \cdot 10^9 \text{ mm}^4$$

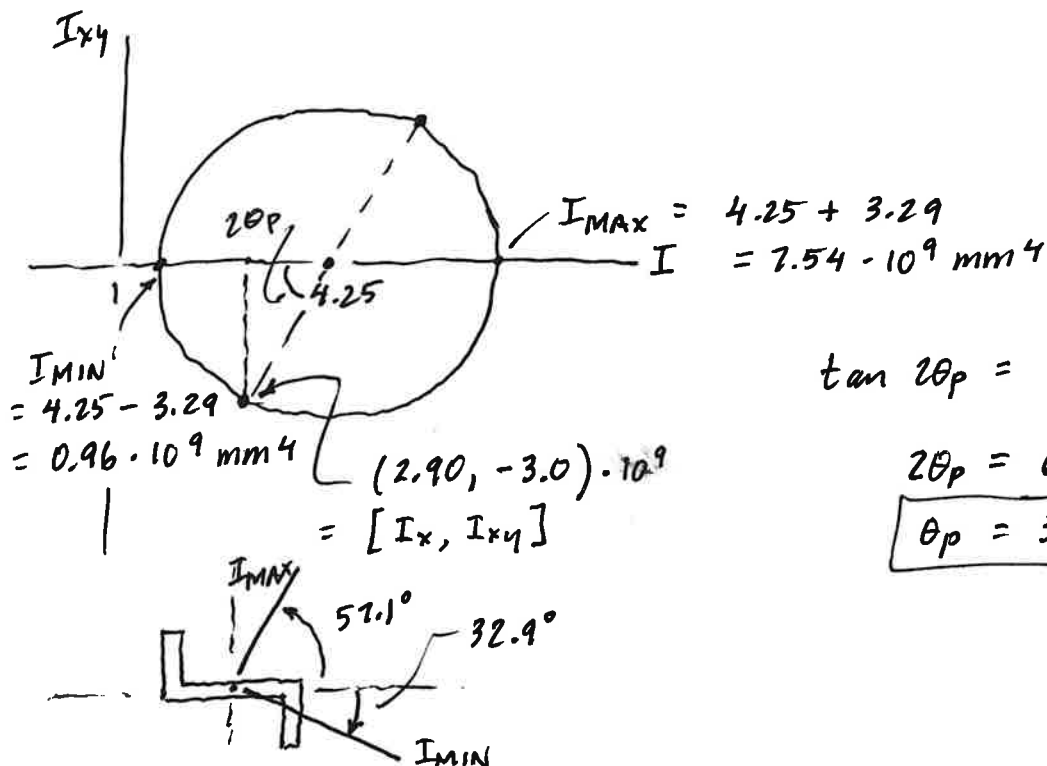
$$I_y = \sum (\bar{I} + Ad^2) = \frac{1}{12} (100)(400)^3 + 2 \left[\frac{1}{12} (400)(100)^3 + (100)(400)(250)^2 \right]$$

$$= 533.33 \cdot 10^6 + 2(2.533 \cdot 10^9) = 5.60 \cdot 10^9 \text{ mm}^4 = I_y$$

$$I_{xy} = \sum (\bar{I}_{xy} + Ad_x d_y) = (400)(100)(+150)(-250) + (400)(100)(+250)(-150) = -3.0 \cdot 10^9 \text{ mm}^4 = I_{xy}$$

Center of circle: $I_{AVG} = \frac{I_x + I_y}{2} = \frac{2.9 + 5.6}{2} = 4.25 \cdot 10^9 \text{ mm}^4$

Radius = $\sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2} = \left[\left(\frac{2.9 - 5.6}{2}\right)^2 + (-3)^2\right]^{1/2} = 3.29 \cdot 10^9 \text{ mm}^4$



$$\tan 2\theta_p = \frac{3}{4.25 - 2.9} = \frac{3}{1.35}$$

$$2\theta_p = 65.8^\circ$$

$$\theta_p = 32.9^\circ$$

This is the angle to the I_{MIN} axis!